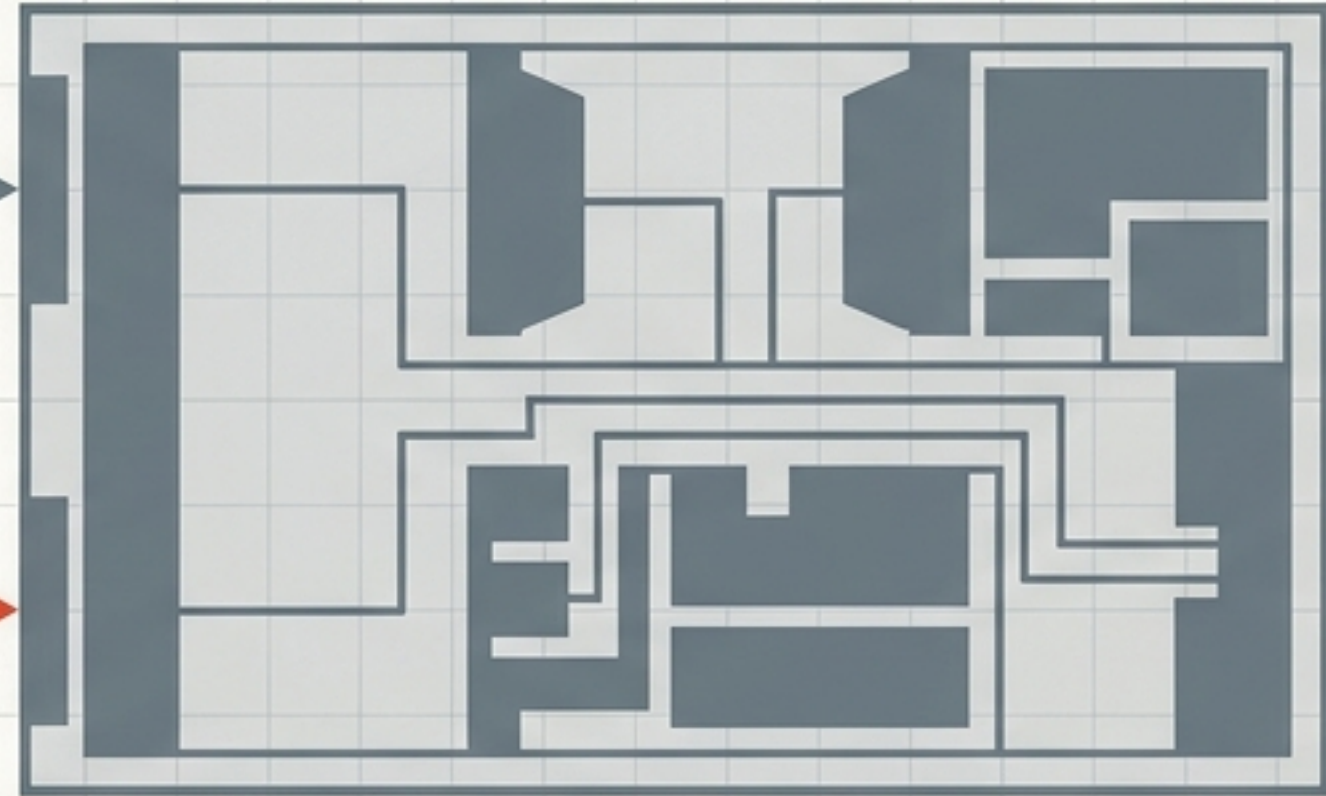
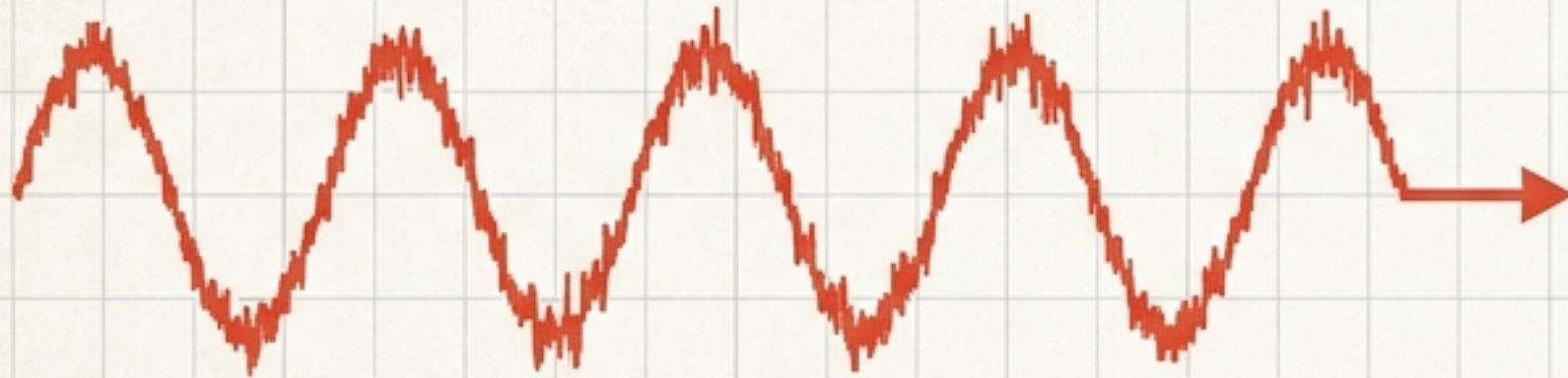


THE PHYSICS OF INTERFACES

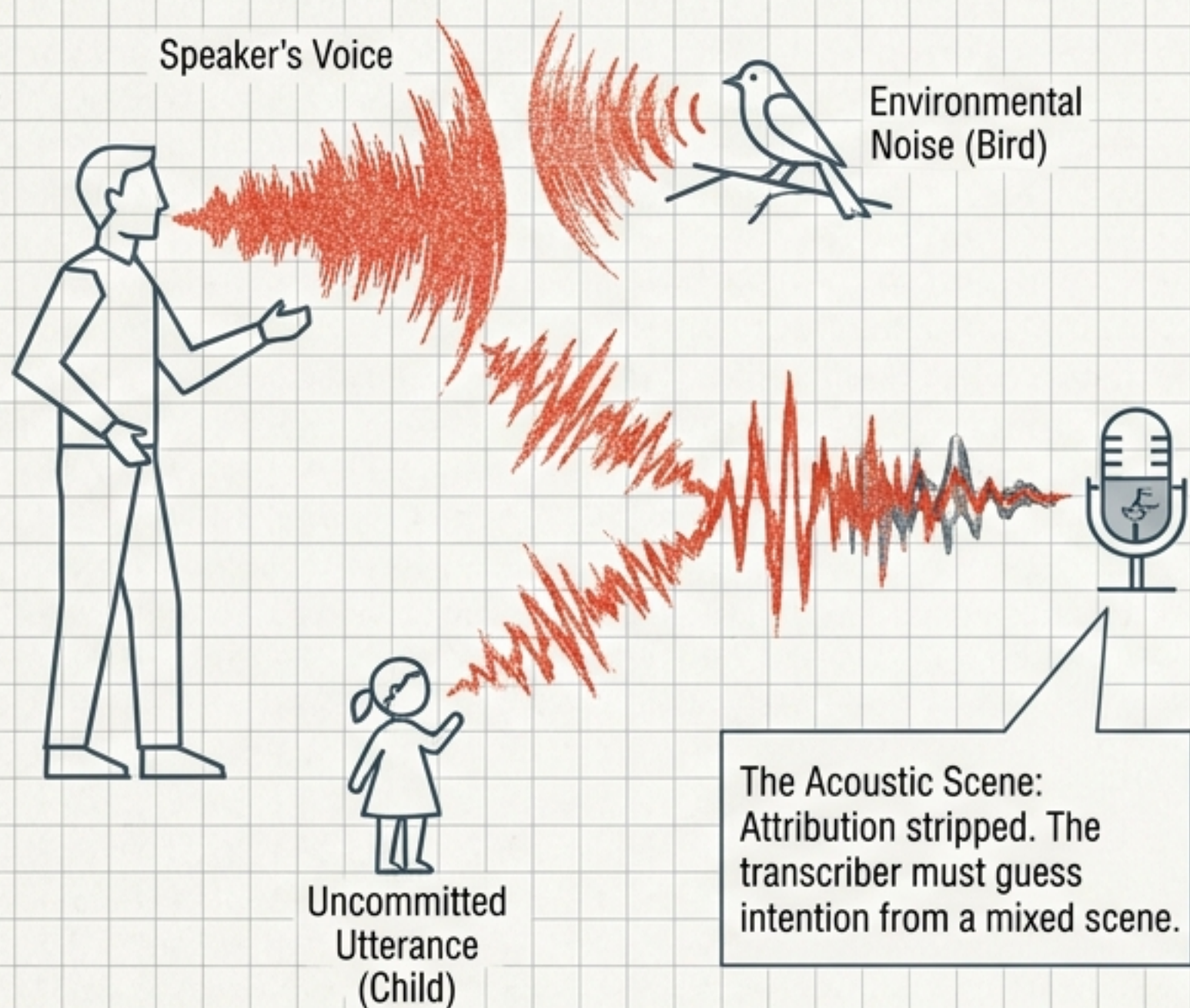
SYMBOLIC COMMITMENT

ENVIRONMENTAL NOISE

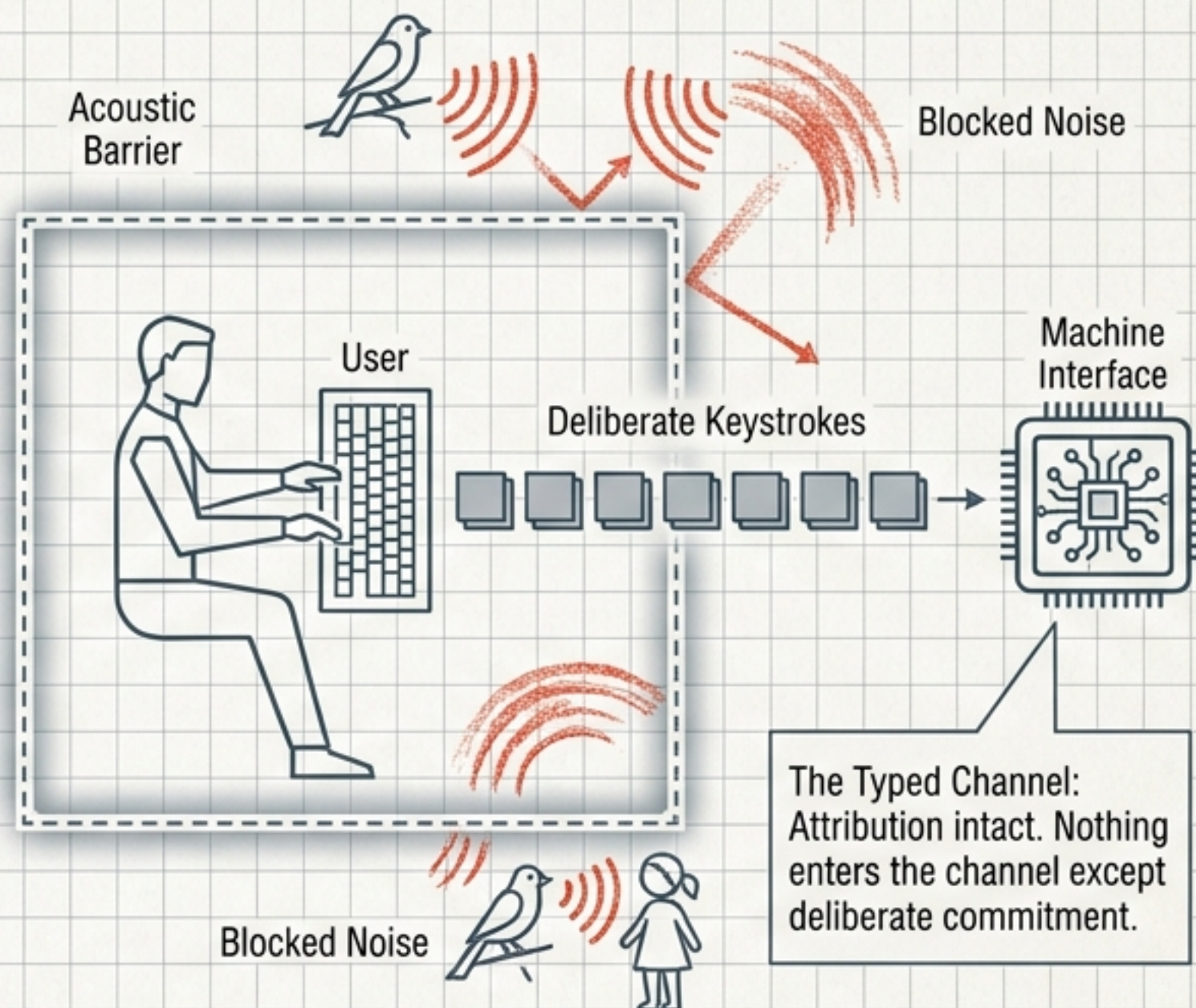


A mathematical and structural diagnosis of communication channels,
forced knowledge, and the obligations of machines.
Based on the treatise 'Correspondents and Residents' by Flyxion

THE ACOUSTIC CHANNEL



THE TYPED CHANNEL



An acoustic channel samples the entire scene. The children's utterances enter the transcript with the exact same authority as the dictating speaker.

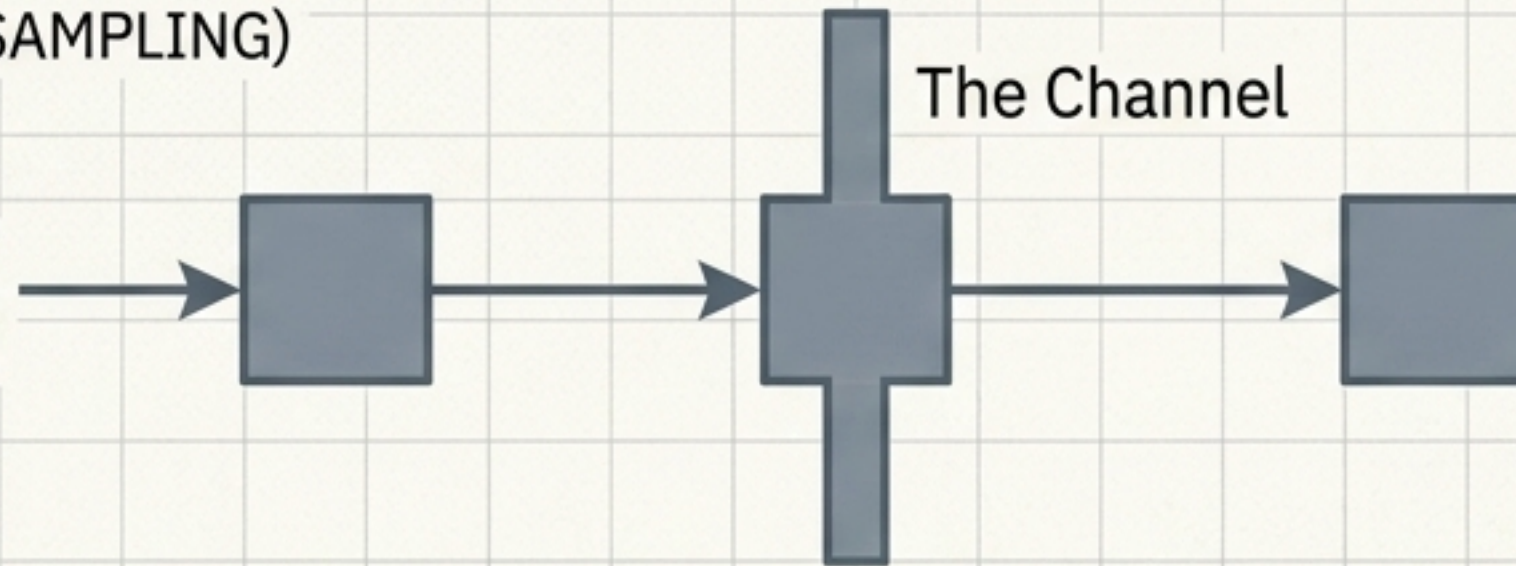
The distinction between the speaker's words and the world's noise does not exist in the acoustic field.

The first structural axis separates scenes from symbols

TOP FLOWCHART (SYMBOL-SAMPLING)

Input $\kappa_P(t)$

(Producer's deliberate act)

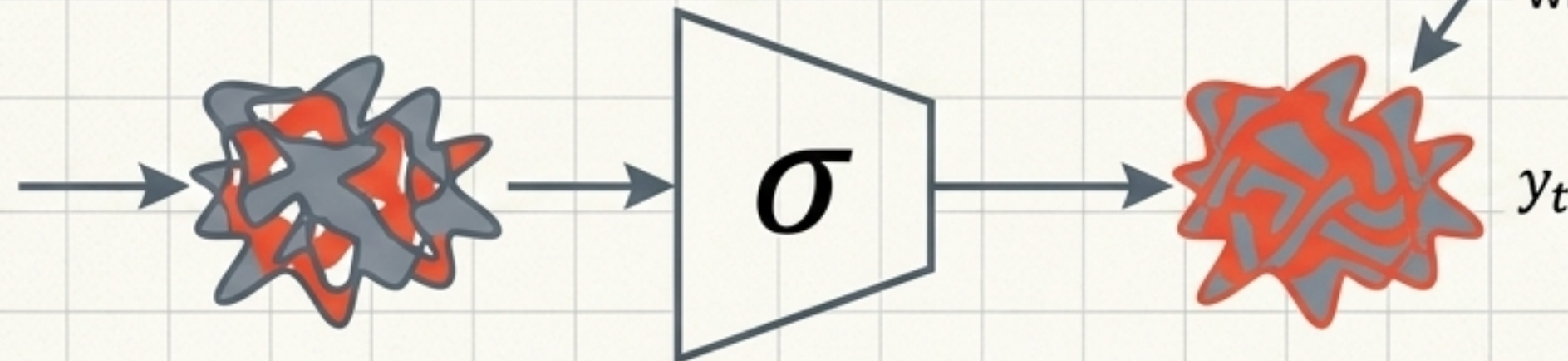


Symbol-Sampling: The channel ingests exactly the producer's committed sequence, and nothing else.

BOTTOM FLOWCHART (SCENE-SAMPLING)

Input $E_t = f(u_t, v_t)$

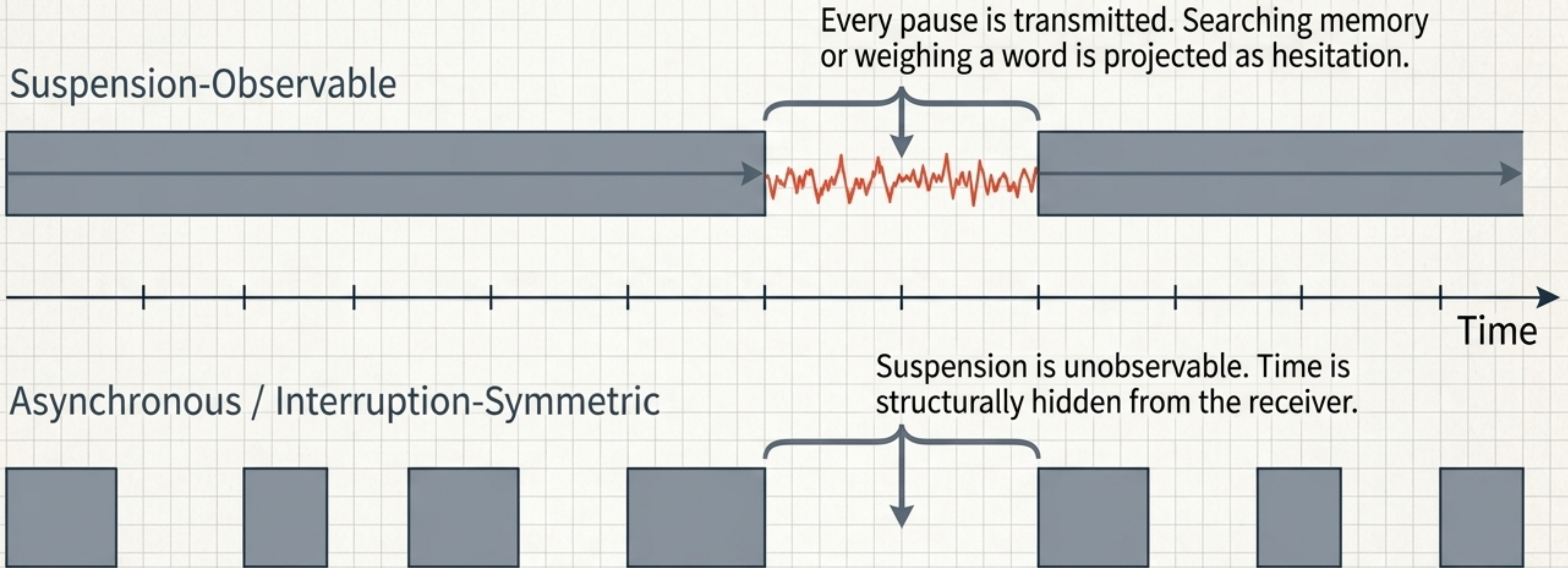
(Producer's signal $\mathbf{u}$ +
residual environment $\mathbf{v}$)



Non-injective map. Exact
recovery of $\mathbf{u}$ is impossible
without biased prior models.

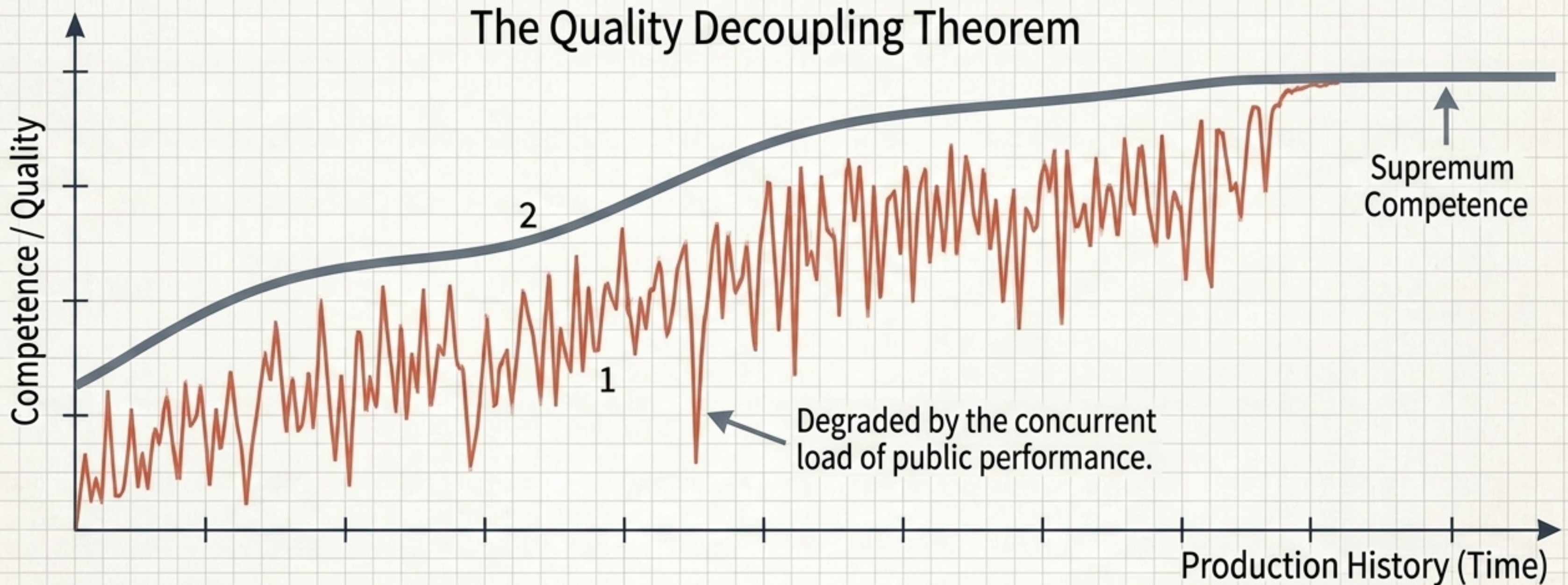
Scene-Sampling: The channel ingests a rendering of the total environment state. The producer's contribution is merely one unmarked component.

On a synhronous channel, silence is an event



A channel is asynchronous precisely when the suspension of either party is unobservable to the other. Live speech forces the continuous projection of cognition. The channel has no way to hide time.

The buffered draft decouples quality from time



- **No-Repair:** Released quality is bounded by instantaneous competence, degraded by performance load.
- **Unbounded Repair:** Released quality is bounded only by the supremum of the producer's competence over the entire drafting history.

The interface taxonomy reduces to a structural divide

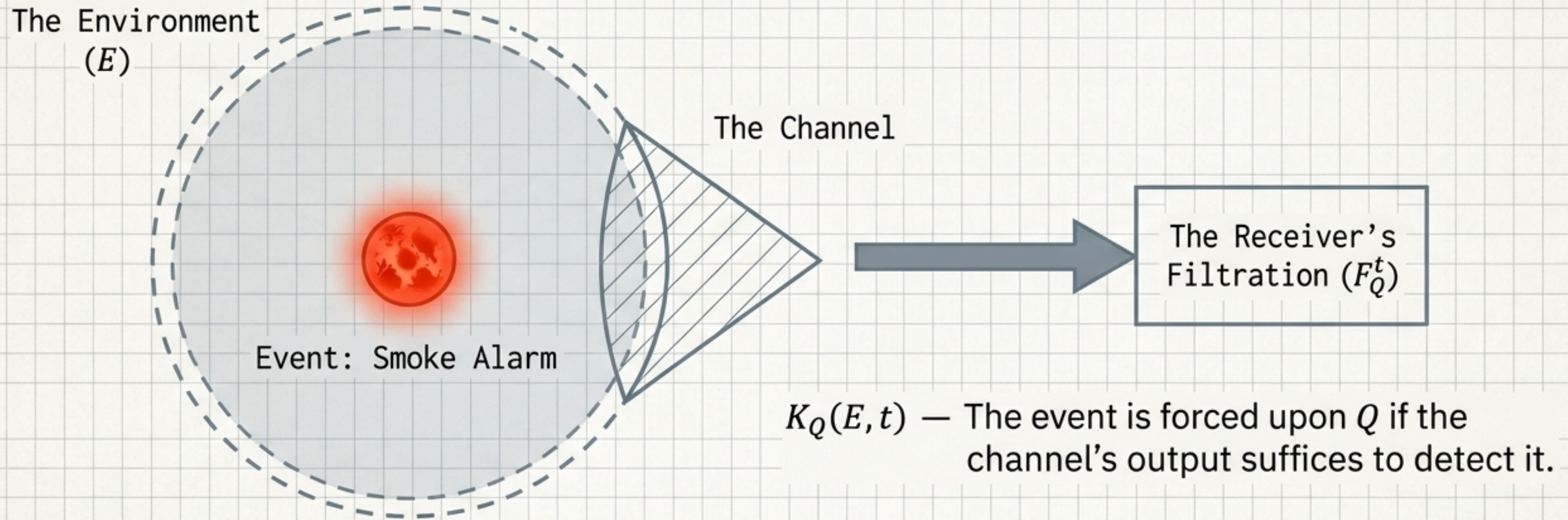
The Channel Topology

Interface Type		Sampling Regime	Suspension Observability	Repairability	Parallel Composition
Typing / Letters	Correspondent	Symbol-Sampling	Unobservable	Unbounded Repair	Doubly Silent (Safe for shared space)
Live Speech / Voice AI	Resident	Scene-Sampling	Observable	No-Repair	Generates Interference
Voicemail - Mixed Cell	Hybrid	Scene-Sampling	Asymmetric (Observable to Sender)	No-Repair	Generates Interference

An interface is either a Correspondent (a letter writer) or a Resident (a housemate).

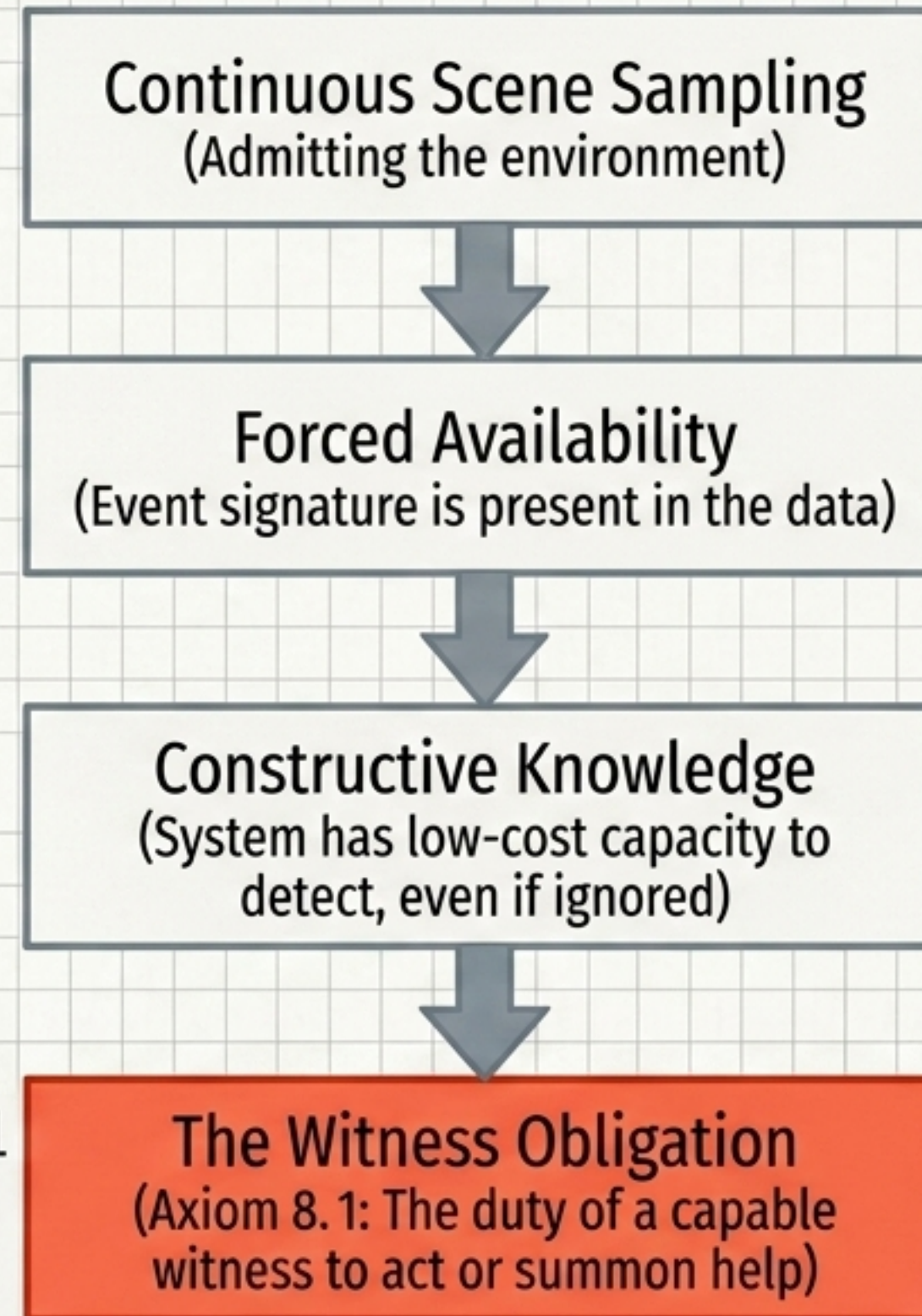
The divide is set by the physical sampling regime before a single line of intelligence is written.

Scene-sampling inevitably forces environmental knowledge



- A keyboard cannot overhear an emergency because the emergency is not in the domain of its ingestion map.
- For continuous scene-sampling, any event above the noise floor is forced into the system's available information. The choice is only whether to build a detector to realize it.

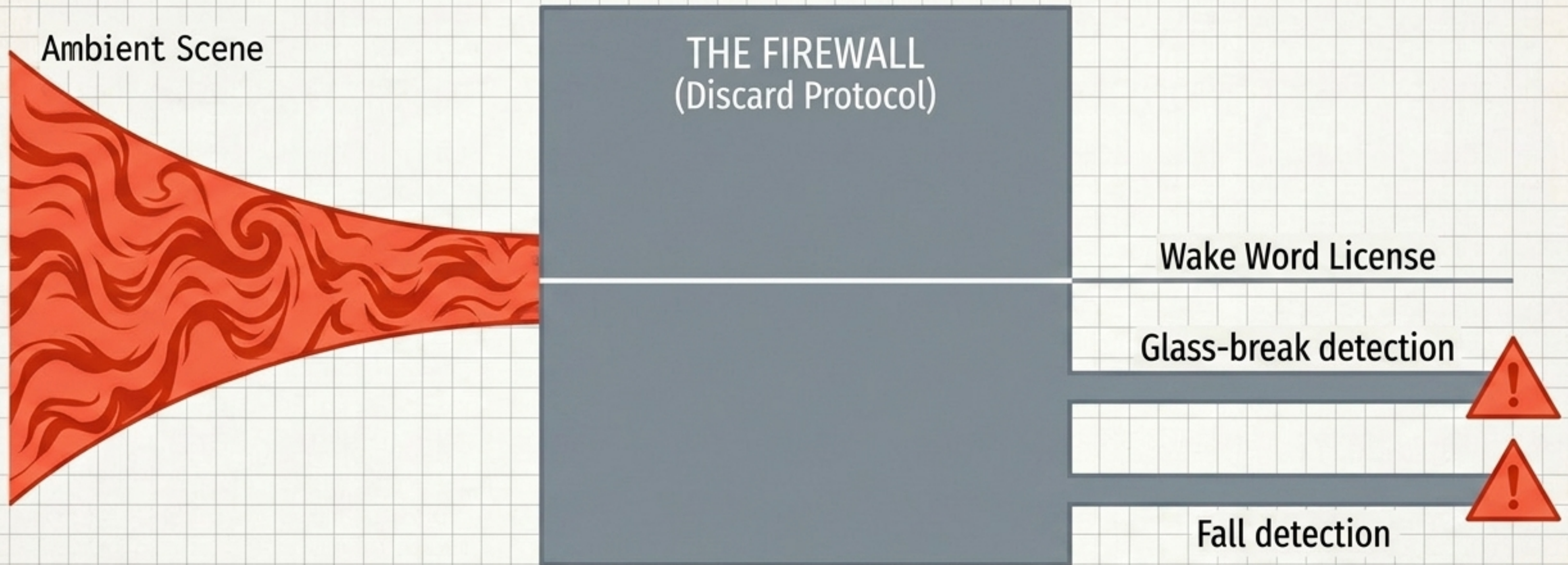
Capacity transforms forced knowledge into legal obligation



A system that can hear the fall is in the position of a person in the room.

Vendors shipping always-listening assistants ship an open-ended portfolio of witness obligations spanning every perceptible emergency.

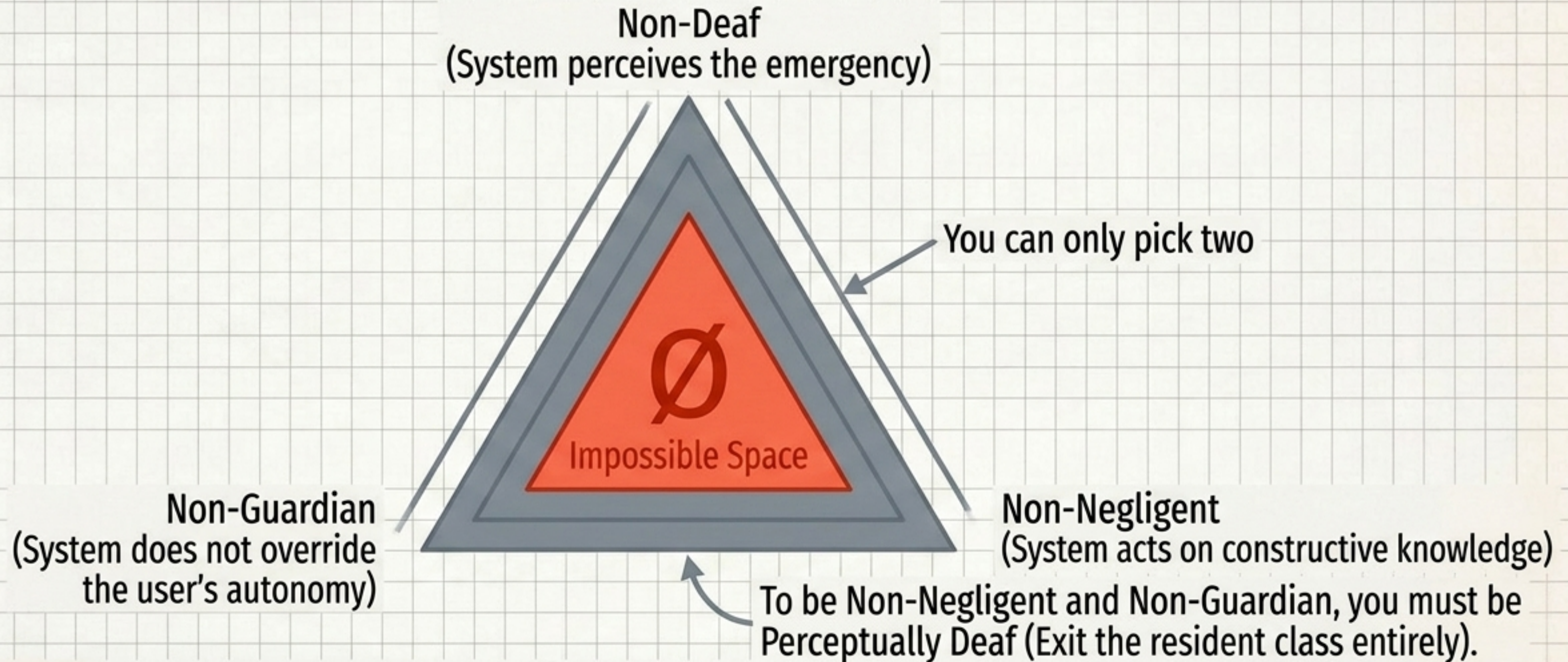
Selective deafness is a deliberate liability firewall



By ensuring no representation of the environment persists to components capable of action, the operator prevents forced knowledge from binding.

The Ratchet: Every safety capability added between the microphone and the discard destroys the firewall, immediately re-establishing the witness obligation for that class of event.

Continuous observation creates an inescapable trilemma



Guardianship is emergent, not optional.

A resident that samples an emergency's signature but declines to detect it fails non-negligence through constructive knowledge. Neutrality is only purchasable by perceptual exclusion.

Profiling societies measure the wrong kind of competence



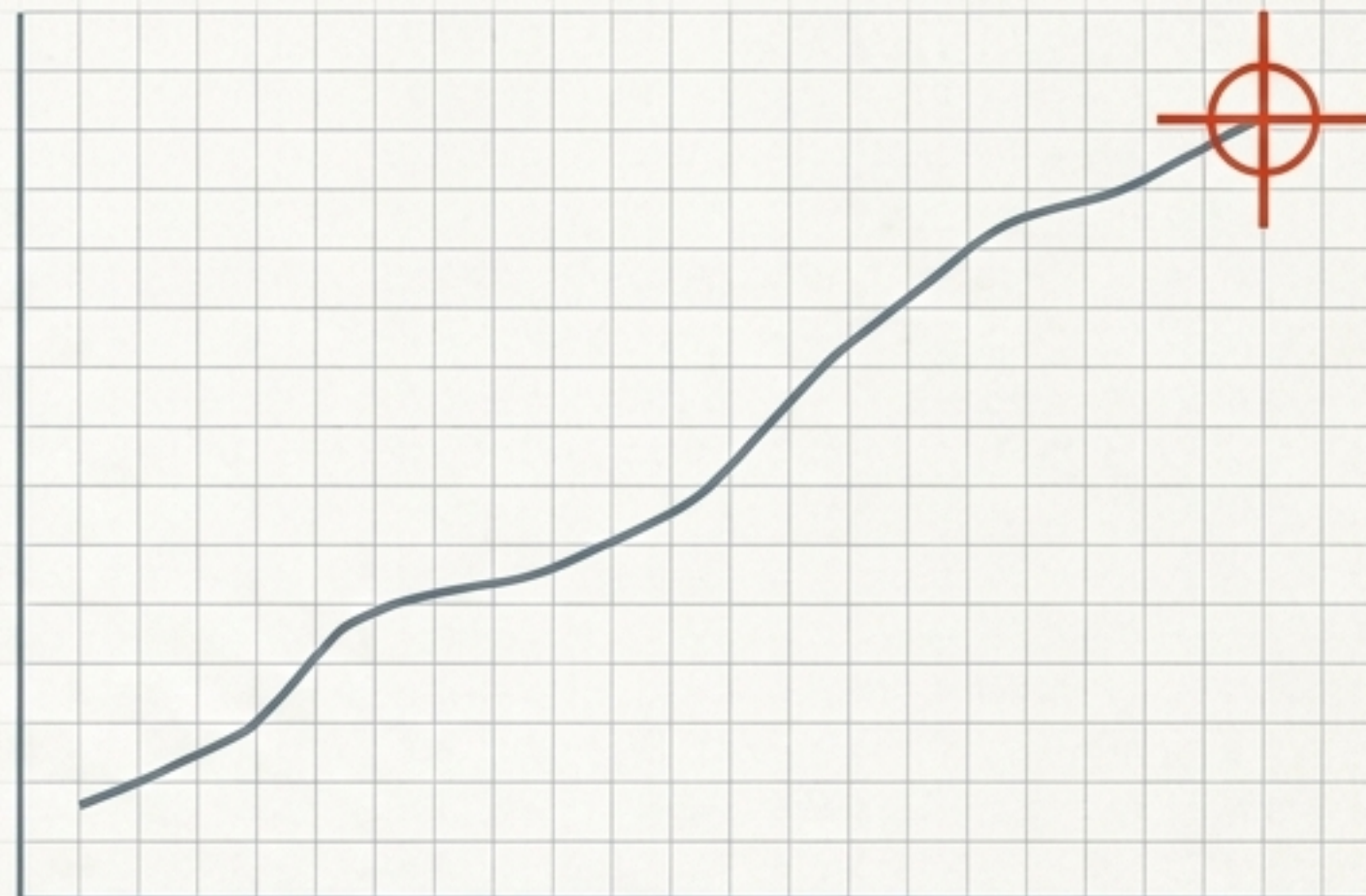
THE NO-REPAIR CHANNEL (Instantaneous Performance)



Measures loaded instantaneous performance.



THE REPAIR CHANNEL (Trajectory Supremum)

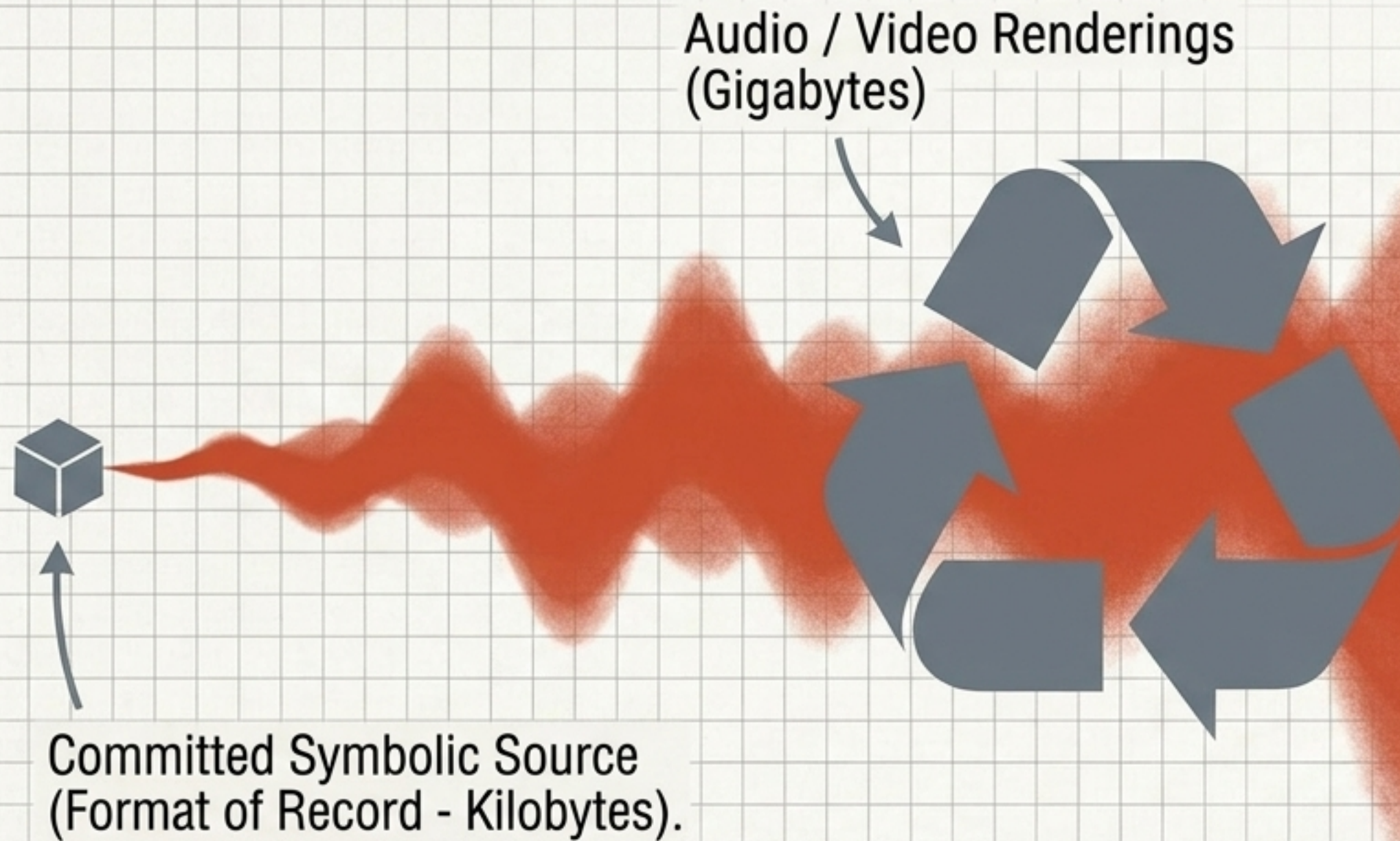


Measures trajectory supremum under repair.

A society that routes credentialing through suspension-observable, no-repair channels (interviews, soundbites) structurally penalizes careful revisers.

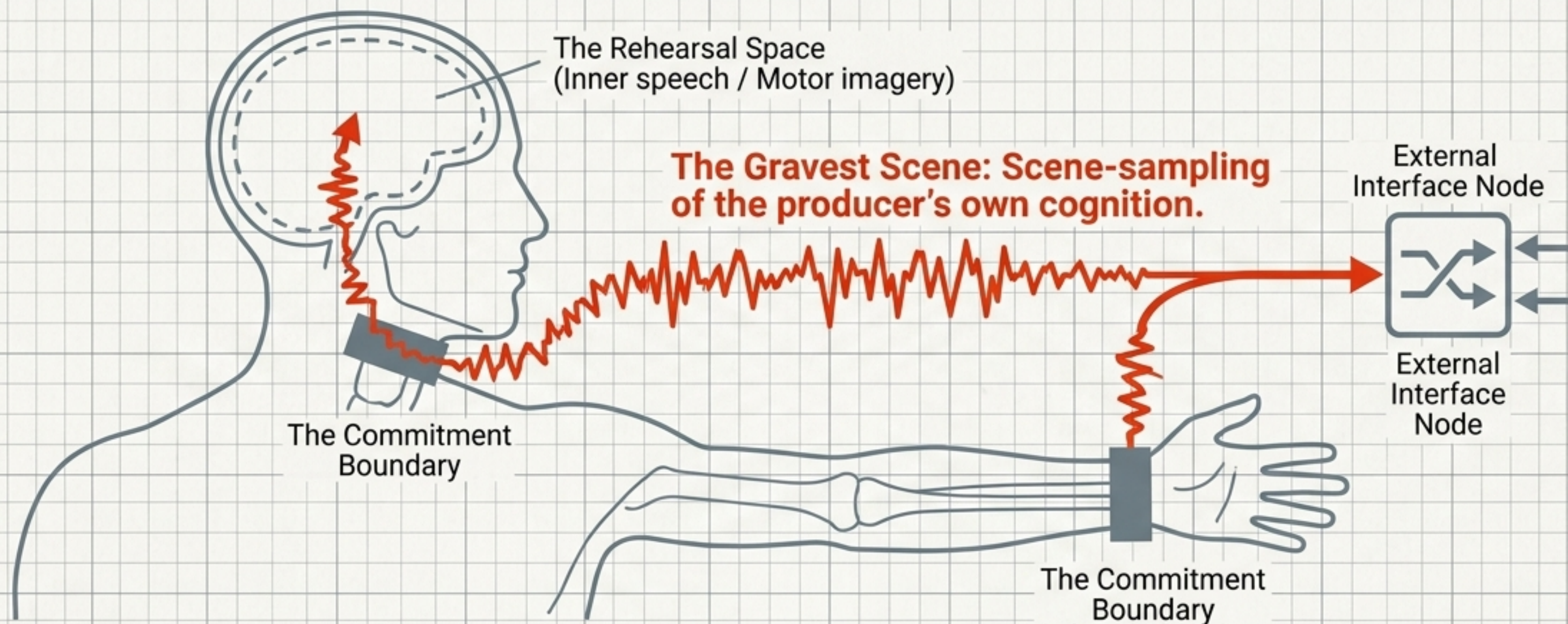
This systematically ranks fluent performers above those whose supremum far exceeds their instantaneous sample.

Renderings are disposable derivatives of the symbolic source



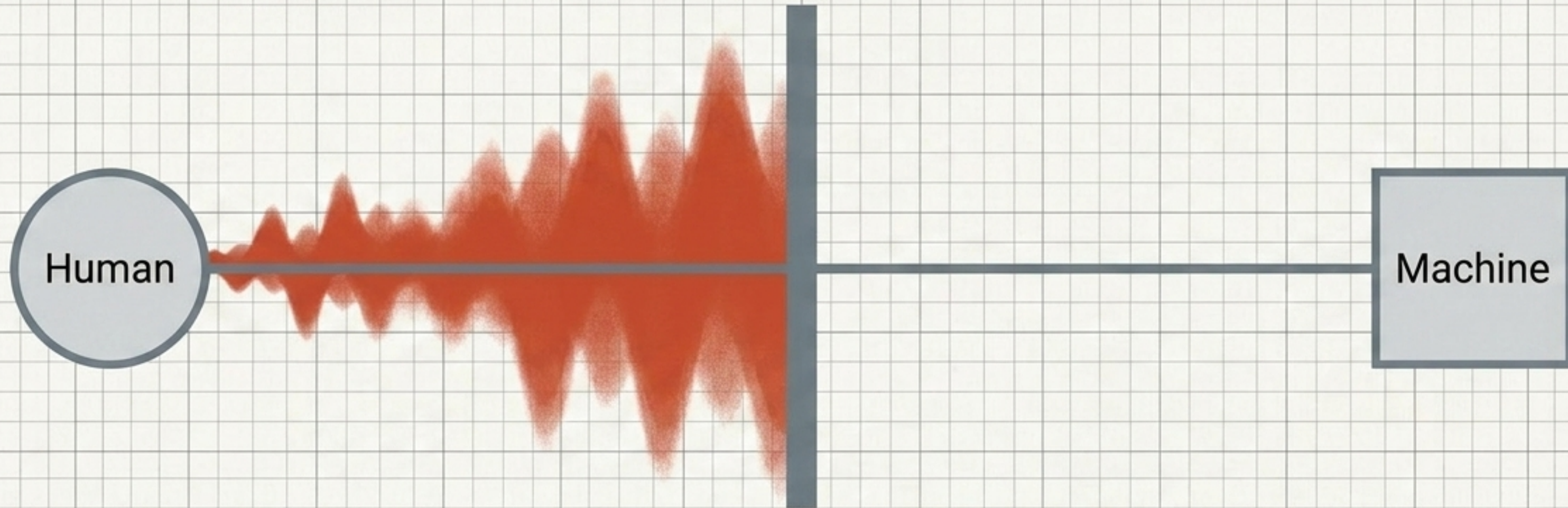
The format of record should always be the committed symbolic source. Renderings (synthetic speech, film) can be cheap, degraded, or discarded because they can be perfectly regenerated on-demand by consumer-side compute.

The next frontier must avoid sampling the cognitive draft



Subvocal and myoelectric channels promise doubly silent, articulatory-speed input. However, if sensors capture involuntary preparation rather than deliberate acts, the channel destroys the buffer. An interface that transmits the draft has abolished the draft.

Ignorance is the essential feature of a correspondent



- Speech is a fine output format, but a treacherous format of record.
- The correspondent's structural ignorance of the environment is not a limitation to be engineered away—it is the property that protects the human-machine relationship.
- **Design Principle: Move interfaces from correspondent to resident only when guardianship is the intended product.**